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Incentive Redesign to Enhance the Capacity of Farmers Practicing Sustainable, Environmentally Friendly Rice Farming in Ngawi Regency, Indonesia

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Abstract: As one of the food baskets in East Java Province, Ngawi Regency plays a strategic role in supporting regional and national food security. The local government has launched the Ngawi Go Organic 2030 program to transform the production system from conventional to organic agriculture. The provision of incentives under this program has not yet served as a lever for rice farmers to aggressively transition to organic agriculture. Only approximately 37% of the identified rice paddy area is currently under Sustainable Environmentally Friendly Agriculture (Pertanian Ramah Lingkungan Berkelanjutan, PRLB) implementation. This study aims to formulate an incentive redesign to enhance the capacity of rice farmers within the PRLB program in Ngawi Regency. The study employs a qualitative method using the Participatory Rural Appraisal (PRA) approach and in-depth interviews. Research informants included personnel from the Ngawi Regency Food Security and Agriculture Office, field agricultural extension workers, PRLB farmer groups, organic farming associations, and marketing institutions. The data were analysed using thematic analysis, while source and methodological triangulation were used to enhance data validity. The results indicate that government incentives have supported the implementation of PRLB; however, gaps remain in marketing, economic benefits, risk management, the use of environmentally friendly inputs, and farmer capacity building. The proposed incentive redesign includes financial and non-financial incentives supported by enhanced technical, managerial, institutional, and digital capacities for farmers. The integration of these components is expected to increase the adoption of PRLB, strengthen farmers' competitiveness and income, and support environmental sustainability in Ngawi Regency.

Keywords: Subsidies; Organic rice; Organic certification; Financial incentives

1. Introduction

Agricultural systems continue to evolve in response to changing societal needs, technological advancements, and increasing demands for high-quality food products. The agricultural sector serves as the backbone for producing sufficient quantities of high-quality food and is expected to ensure the sustainability of natural resources (Puji et al., 2022). Demand for sustainably produced food continues to rise as consumers pay increasing attention to health, food safety, and environmental concerns (Timpanaro & Cascone, 2022).

Ngawi Regency is one of the main rice-producing centers in East Java Province, with an estimated production of 772,571 metric tons in 2025 (Badan Pusat Statistik Provinsi Jawa Timur, 2025). The Ngawi Regency government has launched the Ngawi Go Organic 2030 program (Irianto et al., 2024) to drive the transition toward a more sustainable production system by developing Sustainable Environmentally Friendly Agriculture (Pertanian Ramah Lingkungan Berkelanjutan, PRLB). PRLB is an environmentally friendly farming system that encourages

increased use of organic fertilizers and a gradual reduction in the use of inorganic fertilizers. The development of PRLB in Ngawi Regency has not yet yielded optimal results. Of the total rice paddy area of 49,622 hectares, 18,381 hectares (approximately 37%) were under PRLB implementation in 2024 (Ministry of Agriculture of the Republic of Indonesia, 2025).

The low adoption rate of sustainable agriculture indicates the presence of barriers to PRLB development that extend beyond technical cultivation aspects (Widiyanti et al., 2024). Farmers still face various challenges, such as uncertainty regarding crop yields during the transition from conventional to environmentally friendly agriculture (Jahantab et al., 2023), limited market access for environmentally friendly products, poor access to information (Walaela et al., 2025), reluctance to adopt complex innovations, and stricter cultivation standards compared to conventional systems (Rizzo et al., 2024). Further, the economic benefits received by farmers are often not commensurate with the risks and investment costs incurred in adopting sustainable agriculture (Varyvoda et al., 2025). These conditions result in a relatively low willingness to switch to environmentally friendly farming systems.

Incentives are a key tool for encouraging changes in farmers' behaviour. Several studies indicate that economic incentives, institutional support, access to financing, technical assistance, and market access can increase the adoption of sustainable agricultural practices (Barbosa, 2024; Dzakiroh et al., 2021; Mapanje et al., 2023). Sustainable agriculture programs generally combine financial and non-financial incentives as a strategy to accelerate the adoption of innovations while strengthening farmers' capacity to implement more environmentally friendly farming practices (van Asseldonk et al., 2023).

The PRLB program is supported through various government interventions and incentive mechanisms, including the implementation of clear technical standards, adequate financial assistance and budget allocations, facilitation of organic certification, and agricultural extension services. The government of Ngawi Regency has implemented various incentive and support programs, yet the adoption rate remains low (Irianto et al., 2025). The effectiveness of incentives depends heavily on their alignment with farmers' needs and the characteristics of sustainable agriculture adoption from economic, social, institutional, and environmental perspectives. Economic incentives, extension services, and institutional support are essential drivers of farmers' adoption of sustainable agricultural practices (Piñeiro et al., 2020). The success of sustainable agriculture adoption is largely determined by farmers' internal characteristics, including digital literacy, individual values, and regional geographic conditions (Ohashi et al., 2024). A one-size-fits-all policy intervention will not be effective across all regions. Most of these studies still focus on identifying factors that influence the adoption of sustainable agriculture in general, while research on redesigning incentives to meet farmers' needs at the regional level remains relatively limited. Therefore, this study was conducted to formulate a redesign of incentives that better align with farmers' needs and regional characteristics to increase the adoption rate of sustainable agriculture.

2. Methodology

This study employed a qualitative method using the Participatory Rural Appraisal (PRA) approach and in-depth interviews. The PRA approach was used to explore the conditions under which activities were implemented in a participatory manner (Fleming et al., 2021), while in-depth interviews were used to obtain detailed information from stakeholders involved in the development of the PRLB in Ngawi Regency (Sarib & Rasak, 2022). The interviews were conducted using a semi-structured interview guide with open-ended questions, enabling informants to explain their experiences, views, and suggestions in greater depth. The topics explored included the implementation of the PRLB program, the types of incentives received, implementation challenges, market conditions, farmers' capacity, development needs, and recommendations for improving the incentive system. Each interview lasted 2–2.5 hours, allowing for an in-depth exploration of information from each informant.

The research was conducted in Ngawi Regency, East Java Province, from March to June 2026. The research location was selected purposively in Ngawi Regency, East Java Province, because it is one of the major rice-producing centers currently developing the Ngawi Go Organic 2030 program through PRLB implementation (Widiyanti et al., 2024). Despite the program implementation, the area adopting PRLB has not increased significantly. Data collection was carried out through participatory observation of PRLB cultivation practices, field notes, a review of program documents, and discussions with stakeholders as part of the PRA process. The results of these observations and discussions strengthened understanding of the actual conditions of PRLB implementation in the field.

Informants were selected using purposive sampling, taking into account their direct involvement in the planning, implementation, facilitation, certification, institutional framework, and marketing of PRLB. This approach was used to obtain accurate and in-depth information (Triyono et al., 2025). A total of 48 key informants were involved, as shown in Table 1. This diversity of informants enabled the study to obtain perspectives from all the key actors involved in the development of PRLB in Ngawi Regency.

During the interview process, the researcher first took field notes on key information, the informants' responses, and findings deemed relevant to the research objectives. To ensure data completeness, the entire interview process

was recorded using an audio recorder after obtaining the informants' consent. The audio recordings were used as backup data to ensure that no important information was left out during the field note-taking process. The interview recordings were then transcribed manually by listening to the recordings repeatedly whilst typing out the entire conversation from start to finish. Once the transcription process was complete, the transcripts were cross-checked against the audio recordings and field notes to ensure the accuracy, completeness and consistency of the data obtained.

Table 1. Research key informants

Informant Category	Positions/Representatives	Number (Person)
Food Security and Agriculture Department of Ngawi Regency	Head of the Department	1
	Head of the Food Crops Production Division	1
	Head of the Food Crops Production Facilities and Infrastructure Division	1
Field Agricultural Extension Workers (Penyuluh Pertanian Lapangan, PPL)	Representatives from each district	19
Chair of the Sustainable Environmentally Friendly Agriculture (Pertanian Ramah Lingkungan Berkelanjutan, PRLB) Group	Representatives from each district	19
Ngawi Organic Center Community (Komunitas Ngawi Organik Center, KNOC)	Chairperson and Secretary	2
Lawu Organic Certification (LOC)	Chairperson and Secretary	2
Kampoeng Beta	Chairperson and Secretary	2
Perseroan Terbatas (PT) Sumber Bhakti	Managing Director	1
Total		48

The data were analysed qualitatively using a thematic analysis approach. The analysis process began with all researchers reading the interview transcripts repeatedly to gain a comprehensive understanding of the information provided by the informants. Coding was carried out manually using pre-defined categories based on the interview guidelines, covering the implementation of the PRLB, the forms of incentives received, implementation constraints, marketing conditions, farmers' capacity, development needs, and recommendations for improving the incentive system.

Subsequently, informants' statements sharing similar meanings or information were grouped under the same code. Related codes were then grouped into broader themes or findings. For example, statements regarding difficulties in finding buyers, limited market information, and dependence on middlemen were grouped as findings concerning limited market access. The results of this grouping were then compared across stakeholder groups to identify similarities, differences and emerging patterns of information.

The coding and interpretation of data were carried out manually by all researchers without the use of qualitative data analysis software. Where there were differences in the coding or interpretation of informants' information, these were discussed by all researchers until a consensus was reached. The final themes were used as the basis for compiling the empirical findings and identifying gaps in the existing PRLB incentive system.

Data validity was assessed through source triangulation and methodological triangulation by comparing information obtained from different stakeholder groups and across interviews, field observations, field notes, and supporting documents, thereby enhancing the credibility and reliability of the findings (Arias Valencia, 2022). Data collection was carried out until data saturation was reached, that is, when the information obtained began to show consistent patterns, no new themes or substantive information were found, and the answers provided by informants tended to be repetitive.

3. Results and Discussion

3.1 Current Government Incentives

The research findings indicate that the Ngawi Regency Government has provided various forms of incentives to support the development of PRLB. These incentives include farmer mentoring and capacity building, assistance with production inputs, provision of agricultural tools and machinery, agricultural insurance, access to finance through the People's Business Credit (Kredit Usaha Rakyat, KUR) scheme, and support through village funds. These forms of incentives not only demonstrate the government's support for the technical aspects of farming but also cover financing and the strengthening of agricultural activities at the village level. A summary of the findings regarding the forms of government incentives, together with statements from the informants, is presented in Table 2.

Table 2 shows that one form of government support experienced by farmers is capacity building through extension activities and field schools. Extension programs and field schools are among the most common forms

of incentives for farmers. Extension activities and field schools serve not only as a medium for technology transfer but also as a means of providing guidance in implementing sustainable, environmentally friendly farming practices (Jabbar et al., 2022). Through these activities, farmers gain knowledge regarding the use of organic fertilizers, soil fertility management, integrated pest management, and the application of PRLB operational standards.

Table 2. Interview findings regarding government incentives in PRLB development

Findings	Informants	Informant Quotations
Farming guidance	Field Agricultural Extension Worker	Support is provided through extension program and field schools, particularly regarding PRLB cultivation and the use of organic fertilizers.
	Chair of the PRLB Group	We are often given the opportunity to learn about PRLB from field extension officers through field extension program and field schools.
Provision of production inputs	Head of the Food Crop Production Facilities and Infrastructure Division	The government provides subsidies for inorganic fertilizers and assistance with organic fertilizers such as Petroganik to support PRLB.
	Chair of the PRLB Group	We continue to receive fertilizer assistance and subsidies from the government for both organic and inorganic fertilizers.
Provision of agricultural tools and machinery	Head of the Food Crop Production Facilities and Infrastructure Division	The government provides agricultural tools and machinery to support PRLB cultivation.
	Chair of the PRLB Group	We have received tools, but we still have to provide some equipment, such as water pumps, ourselves.
Agricultural insurance	Head of the Food Crop Production Division	The government subsidises insurance premiums so that the costs borne by farmers are reduced.
	Chair of the PRLB Group	With these subsidies, premiums become more affordable, and farmers are free to choose whether or not to participate.
Access to People's Business Credit (KUR)	Head of the Department	The government is currently also offering loans to people who lack capital.
Village fund support	Head of the Department	Village funds are channelled towards training in organic fertilizers, biological pesticides, and PRLB demonstration projects.
	Field Agricultural Extension Worker	Village funds support training and agricultural practices that are more environmentally friendly.

Note: PRLB = Sustainable Environmentally Friendly Agriculture (Pertanian Ramah Lingkungan Berkelanjutan); KUR = People's Business Credit (Kredit Usaha Rakyat). Source: Authors' analysis of interview data (2026).

The findings in Table 2 also indicate that the government provides direct support for production inputs. In each planting season, the government provides inorganic fertilizer subsidies to farmers based on the size of their rice-farming land holdings. The purchase of these subsidized fertilizers is made using a farmer's card at agricultural kiosks designated as official distributors (Wahyudi et al., 2021). The government also provides organic fertilizer assistance in the form of Petroganik, though the amount is lower than that for inorganic fertilizers. These fertilizer subsidies are expected to increase production and reduce production costs (Pauw, 2022).

Government support is also provided in the form of agricultural tools and machinery, as stated by the informants in Table 2. The availability of farming tools and machinery is deemed capable of improving farming efficiency and reducing crop yield losses (Rusmono et al., 2026). However, the interview findings also indicate that the agricultural tools and machinery provided do not yet fully meet farmers' equipment needs. The farmers still need to purchase supporting equipment such as water pumps independently. Water pumps are generally used during the planting season, at the beginning of the dry season, and throughout the dry season to ensure irrigation water is available.

The government also provides support in the form of access to finance through the KUR scheme. According to the Head of the Department in Table 2, the KUR program provides access to finance for individuals with limited capital, including those engaged in agricultural activities. This financial support complements the provision of production inputs by offering farmers an alternative source of capital for running their farming businesses.

Another form of support identified in the study is protection against farming risks through an agricultural insurance program. The premium that farmers pay is currently IDR36,000 per hectare due to the subsidy. Without the subsidy, the premium to be paid is approximately IDR180,000 per hectare per growing season. The premium payment system is voluntary, as farmers can decide for themselves whether to participate based on the level of risk they face (Brouwer et al., 2025). If farmers perceive the upcoming growing season to be high-risk, they typically enrol in the insurance program; however, if the risk is deemed low, they do not pay the insurance premium. Thus, premium payments can be activated or suspended as needed.

As shown in the statements from the Head of the Department and the Field Extension Officer in Table 2, the

development of PRLB is also supported by village fund allocations. According to the research findings, village governments in Ngawi Regency allocate approximately IDR25–30 million annually to support PRLB activities. These village funds are used to train farmer groups in the production of organic fertilizers or biological pesticides and to conduct demonstration plots for their application (Irianto et al., 2024). The expected outcomes of these activities are to accelerate the transformation toward sustainable, environmentally friendly agriculture (Irianto et al., 2024). Although there are currently budget adjustments due to the "*Merah Putih*" village cooperative program, the regency government continues to instruct villages to allocate funds for PRLB activities.

These findings indicate that the Ngawi Regency Government has provided various forms of incentives, including capacity building, assistance with production facilities, financing, and institutional support. These diverse incentives provide a crucial foundation for promoting PRLB development in Ngawi Regency.

3.2 Evaluation of the Current State of Sustainable Environmentally Friendly Agriculture

3.2.1 Farmers' bargaining position

The research findings indicate that the bargaining position of PRLB farmers in the marketing of agricultural produce remains relatively weak. This situation is evident from limited access to markets and the absorption of produce, farmer organisations that remain more focused on cultivation activities, dependence on middlemen and local markets, and the lack of strong marketing partnerships with business actors. These findings suggest that the marketing challenges faced by PRLB farmers are not only related to market availability, but also to the capacity of farmers and farmers' organisations to set prices, manage harvests, and build more sustainable marketing relationships. A summary of the interview findings regarding the bargaining power of PRLB farmers is presented in Table 3.

Table 3. Bargaining power of PRLB farmers in the marketing of agricultural produce

Findings	Informants	Informant Quotations
Limited market access and market uptake	Chair of the PRLB Group	We are still finding it difficult to set prices because sales are still dependent on middlemen, so prices are dictated by them.
		We are not yet able to market the entire harvest ourselves.
	Managing Director of PT Sumber Bhakti	We are not yet able to absorb the full PRLB rice yield because the available market is still limited.
Institutions focus more on cultivation	Chair of the PRLB Group	Up to now, the farmers' group has mainly focused on cultivation activities and the farmers' needs. As for marketing, we still sell a great deal through our usual middlemen.
	Field Agricultural Extension Worker	Together with the farmers' group, we have been active in cultivation, but our marketing efforts are not yet very strong.
Dependence on middlemen and local markets	Chair of the PRLB Group	We still sell individually to middlemen or at local markets. There are no regular buyers willing to enter into long-term contracts.
		Every time we harvest, we simply accept whatever price is offered that day; there is no guarantee of a stable price.
Lack of strong marketing partnerships	Chair of the PRLB Group	There have been discussions with companies or supermarkets, but their terms are harsh, and no written agreement has yet been reached.

Note: PRLB = Sustainable Environmentally Friendly Agriculture (Pertanian Ramah Lingkungan Berkelanjutan). Source: Authors' analysis of interview data (2026).

Table 3 shows that limited market access and the uptake of produce are among the factors weakening the bargaining position of PRLB farmers. Market access remains a challenge for PRLB farmers. Farmers have limited ability to set selling prices and still depend on middlemen or local buyers. Market access is one of the factors that influence the economic benefits farmers receive from a farming system (Aku et al., 2018). Regional companies responsible for purchasing rice from organic or PRLB farmers have not been able to absorb the entire production and offer competitive prices. This has led farmers to be reluctant to adopt PRLB farming because the selling price of paddy or rice is not significantly different from that of conventional farming (Widiyanti et al., 2024). Work in PRLB farming is far more complex than in conventional farming, thus requiring more resources (Irianto et al., 2024). Market constraints not only affect farmers' ability to sell their produce, but may also reduce the economic incentives to maintain or expand the adoption of PRLB. This finding is also supported by Meemken (2020)'s research, which revealed that while some farmers achieved good economic returns, others saw their incomes decline because institutions were unable to absorb the entire harvest, compounded by rising production costs.

The findings in Table 3 also indicate that farmers' organisations still perform more functions related to cultivation than to marketing. Farmer institutions serve as a means of coordination, information dissemination, and the implementation of PRLB programs. Farmer groups also play a crucial role in implementing various agricultural development programs at the field level (Komariyati et al., 2026). However, the functions of these institutions remain more focused on technical aspects of cultivation than on the marketing of produce. This finding

was also confirmed by a study by Kartika et al. (2026) in Tasikmalaya Regency, where the role of these organisations in marketing remains very weak, particularly in organic rice farming. The abundance of middlemen means farmers have no difficulty selling their paddy, yet they remain merely price takers. Only a small fraction of farmer groups can manage their members' harvests, enabling them to earn higher incomes than members of other farmer groups (Raya, 2014). Farmer groups with an agent of change are better able to manage and optimize their roles (Triyono et al., 2025).

These limitations in bargaining power are further highlighted by the lack of strong marketing partnerships identified in the study. As shown in Table 3, farmers still market their produce individually or through established local marketing networks. Although there have been efforts to establish relationships with companies or supermarkets, requirements deemed onerous and the absence of written agreements have prevented these relationships from developing into long-term partnerships. PRLB farmer groups still face marketing barriers (Widiyanti et al., 2024), as business partnerships between PRLB farmers and business actors or marketing institutions remain relatively limited. PRLB farmers still market their products individually or through existing local marketing networks (Irianto et al., 2023). These conditions indicate that the marketing relationships that have formed remain informal and are not yet supported by long-term cooperation. Consequently, farmers continue to face market and price uncertainties when marketing their harvests. These findings are also consistent with those of Ningrum & Subroto (2020) who demonstrated that market interventions in organic agriculture in Indonesia have not yet significantly strengthened farmers' autonomy.

3.2.2 Market prices for rice produced under Sustainable Environmentally Friendly Agriculture

Research findings indicate that PRLB products have the potential to command higher selling prices than conventional products; however, these benefits have not yet been felt equally by all farmers. Differences in access to price information, markets that offer added value, and the ability to meet certification requirements are some of the factors influencing farmers' ability to secure premium prices. Consequently, the existence of premium prices does not automatically provide the same economic benefits to all PRLB farmers. A summary of the interview findings regarding farmers' access to PRLB premium prices is presented in Table 4.

Table 4. Farmers' access to the PRLB premium price

Findings	Informants	Informant Quotations
Farmers have not yet benefited equally from the PRLB premium price	Chair of the PRLB Group	The price of PRLB rice could actually be higher, but not all farmers are yet able to secure such prices. It is true that some of our friends have received higher prices, but this is not yet the case across the board.
	Field Agricultural Extension Worker	Not all farmers are receiving higher prices because there are still disparities in access to information and suitable markets.
	Chair of the PRLB Group	We still need clearer information regarding PRLB rice prices, particularly if there are markets that do offer special prices, because whilst some have mentioned high prices, we have not yet experienced them ourselves.
	Field Agricultural Extension Worker	The government has introduced a certification scheme; certification can help products gain easier acceptance in certain markets, but not many farmers have obtained certification yet.
	Chair of the PRLB Group	Certification can indeed add value, but the process of obtaining it is very difficult, particularly for our older colleagues, and due to limited capital, so the certification rate remains low and the benefits of higher prices have not yet been felt by all farmers.

Note: PRLB = Sustainable Environmentally Friendly Agriculture (Pertanian Ramah Lingkungan Berkelanjutan). Source: Authors' analysis of interview data (2026).

Based on Table 4, not all farmers are yet able to benefit equally from the premium price for PRLB products. Market prices are one of the factors influencing farmers' interest in adopting PRLB. Research findings indicate that PRLB paddy or rice has the potential to command a higher selling price than conventional rice, although this benefit has not yet been felt equally by all farmers. Differences in the selling price of environmentally friendly products are generally influenced by market access; farmers can command higher prices when they have access to suitable markets (Zhang et al., 2024). Not all farmers or farmer groups can access these markets due to limitations in information, networking, and communication skills during negotiations. In line with Meemken (2020), farmers still face limited access to price information, restricted marketing networks that leave them dependent on a single buyer, and inadequate skills, all of which ultimately hinder their ability to secure premium prices.

The ability to secure a premium price is also linked to market developments for products with health and environmental benefits. Healthier, more environmentally friendly food products are beginning to offer growing market opportunities. However, the market for PRLB rice in Ngawi Regency remains limited, so not all of the

farmers' production can be absorbed by market segments that offer value-added benefits. Increasing consumer awareness of health and environmental issues is one factor driving the development of markets for sustainable products (Sun & Yoon, 2022). The findings in Table 4 also indicate that certification is one of the factors influencing the ability of PRLB products to enter specific markets and achieve added value. This situation is also linked to market recognition of product quality. In practice, certified products generally find it easier to gain consumer trust and gain acceptance in specific market segments (Zhang et al., 2024). Although the government has facilitated certification, the number of certified farmers remains limited, so the resulting economic benefits have not yet been widely felt. These limitations relate not only to the certification process, but also to farmers' ability to meet the stipulated requirements. This finding is consistent with Alexandre de Lima et al. (2021), who showed that organic farming can command a premium price, but these economic benefits are not felt evenly due to differences in farmers' ability to access markets and meet organic standards.

3.3 Identification of Incentive Gaps

The research findings indicate that, although various forms of government support have been provided to facilitate the implementation of PRLB, there are still several farmers' needs that have not been fully accommodated. These needs relate not only to production aspects but also encompass access to market information, economic incentives, protection against risks during the implementation of PRLB, the availability of environmentally friendly inputs, and farmers' managerial capabilities. Findings from interviews regarding these needs and support gaps are presented in Table 5.

Table 5. Farmers' needs and support gaps in the development of PRLB

Findings/Needs	Informants	Informant Quotations
The need for information on market access	Chair of the PRLB Group	We have often received information sessions or training on how to cultivate PRLB, but when it comes to harvest time, we're left to our own devices, so we're actually unsure where to distribute our PRLB produce to get a higher price.
Economic incentives are needed	Chair of the PRLB Group	PRLB produce is sold in exactly the same way as ordinary agricultural produce, without any price premium; this has left me and my colleagues rather disappointed, as the effort required for PRLB is greater but the price remains the same. Furthermore, to secure a higher price, certification is required, which requires additional capital, whilst we have limited funds.
	Managing Director of PT Sumber Bhakti	We are not yet able to absorb the entire PRLB rice harvest; moreover, the low level of certification means the market remains limited, so pricing and marketing continue to be challenges.
Risk protection needs to be strengthened during the implementation of the PRLB	Head of the Food Crops Production Division	We have endeavoured to provide insurance support to farmers, and we have also offered subsidies to make insurance costs more affordable.
	Chair of the PRLB Group	We did receive assistance from the insurance provider, but the claim requirements were quite stringent; the damage or yield loss had to exceed 75%, whereas when my colleagues and I switched to PRLB, our yield losses ranged from 20% to 40%. This is a factor that needs to be considered so that insurance schemes providing coverage during that transition period can be made available.
Support for organic fertilizer inputs needs to be strengthened	Head of the Food Crops Production Division	We have provided subsidies to the community in the form of organic and inorganic fertilizers, but the quantity of organic fertilizer is still limited.
	Chair of the PRLB Group	We have actually received subsidies for both organic and inorganic fertilizers, but it seems odd that, whilst the government is promoting organic farming, the majority of subsidies are for inorganic fertilizers.
	Field Agricultural Extension Worker	We are doing our best to provide training in organic fertilizer production, so that the community can become self-sufficient in this regard, but we still need support from the government.
Farmers need managerial skills	Chair of the PRLB Group	We still need assistance in managing our farming businesses, particularly in business planning and marketing development, as so far, we have only known how to grow crops.

Note: PRLB = Sustainable Environmentally Friendly Agriculture (Pertanian Ramah Lingkungan Berkelanjutan). Source: Authors' analysis of interview data (2026).

According to Table 5, farmers' primary need is for information and guidance in accessing markets. Most PRLB farmers still market their produce through the same channels as conventional products, so the value added they receive is not yet optimal. Market access is one of the factors influencing the economic benefits farmers receive (Aku et al., 2018). This situation highlights the first gap: support for marketing aspects is still not as strong as

support for production aspects. This has also been observed in Poland, where organic farmers identify market, marketing, and institutional barriers as the main impediments to the development of organic agriculture, even though production support is available (Łuczka & Kalinowski, 2020).

The second need highlighted in Table 5 is the need for economic incentives. Opportunities to secure premium prices do exist, but they are not yet accessible to all farmers because they remain dependent on marketing networks and access to specific markets. In addition, farmers need capital support to cover the costs of certification, which may be a prerequisite for gaining access to premium markets. These findings point to a second gap, namely the lack of adequate capital support to help farmers gain equitable access to premium markets. This finding aligns with research conducted across 97 developing countries in Asia, Africa, and Latin America, which showed that the economic benefits of organic and sustainable agricultural standards tend to vary and are not equally enjoyed by all farmers (Meemken, 2020).

Table 5 also shows that farmers need more appropriate risk protection during the implementation of PRLB. The government has provided agricultural insurance subsidies. The insurance provided focuses on crop failure in general and does not yet specifically address the risks associated with the transition from conventional to organic farming. Agricultural insurance is provided if the rate of crop failure or crop damage exceeds 75%. If the damage rate is less than 75%, insurance is not provided. Farmers' decisions to participate in insurance are often influenced by their perceptions of farming risks (Jin et al., 2016). This situation highlights the third gap: that risk management for farming operations during the transition period is not yet fully protected. This finding aligns with Bhatt & John (Bhatt & John, 2023), who noted that during the transition to organic farming, farmers face the risk of reduced yields and income, yet support to protect them from these risks remains limited.

The fourth need relates to support for the use of inputs that are more in line with PRLB principles, particularly organic fertilizers. Subsidies for the use of production inputs are still allocated more heavily to inorganic fertilizers than to organic ones, even though the PRLB encourages reducing the use of chemical inputs. Incentive policies can influence farmers' decisions regarding the use of production inputs (Pan et al., 2022). Farmers still use inorganic fertilizers at recommended doses and combine them with organic fertilizers. This indicates a fourth gap between the PRLB's objectives and the available forms of support. This finding aligns with Kishore et al. (2021), who note that government support for organic fertilizers still faces budget allocation constraints, while fertilizer subsidy policies remain predominantly oriented toward inorganic fertilizers to support increased food production.

The fifth need highlighted in Table 5 is capacity building for farmers' managerial skills. Extension services and field schools have helped improve farmers' understanding of environmentally friendly farming practices; however, the materials provided still focus primarily on technical cultivation aspects. Farmers' capacity is determined not only by technical skills but also by managerial capabilities and strategies for expanding market access (Permadi & Winarti, 2025). These findings point to the fifth gap in farmer capacity building. Farmers' capacity to build market networks or partnership networks remains weak. This finding aligns with Meekaew & Chamaratana (2024), who notes that organic farming training programs generally still focus on the technical aspects of cultivation, while the development of farmers' entrepreneurial competencies, business planning, financial literacy, and managerial skills remains relatively limited.

In general, the available incentives have supported the implementation of PRLB, but they remain more focused on increasing production. Meanwhile, in the field, farmers' current needs also include capacity building in marketing, economic benefits, risk management, the use of environmentally friendly inputs, and business. Aligning incentive design with farmers' needs is crucial in enhancing the effectiveness of sustainable agriculture programs.

3.4 Financial Incentive Redesign

Based on the various gaps identified, adjustments to the incentive design are required so that government support is not only oriented towards increasing production, but is also capable of providing economic benefits and risk protection for farmers during the implementation of PRLB. Recommendations for the redesign of incentives based on field findings are presented in Table 6.

One way in which incentives could be adjusted is by gradually shifting the composition of fertilizer subsidies from inorganic to organic fertilizers. The effectiveness of incentives is greatly influenced by their alignment with the desired behavioural change (Piñeiro et al., 2020). This policy aims to encourage the use of more environmentally friendly inputs while improving soil quality. This approach could involve gradually reducing inorganic fertilizer subsidies while increasing support for organic fertilizers to promote organic farming during the transition period (Liao et al., 2025). Inorganic fertilizer subsidies could be reduced, with the savings redirected to an equivalent increase in organic fertilizer subsidies. The gradual adoption of organic fertilizers can improve soil structure and organic matter content (Liu et al., 2024).

Farmers' capital needs can be addressed by developing a green credit scheme for PRLB farmers. Limited capital is often a barrier to the adoption of innovations and investments in the agricultural sector (Liao et al., 2025). This scheme can be channeled through low-interest financing with more flexible terms for farmers who adopt environmentally friendly farming practices. Support through green credit is expected to accelerate technology

adoption and the development of sustainable farming enterprises (Li et al., 2024). In addition to capital, farmers also need to derive clearer economic benefits from implementing PRLB. This green credit scheme could also be made available to farmer groups applying for new organic certification or certification renewal. Organic certification is essential for organic products marketed in premium markets or for export (Boys et al., 2022).

Table 6. Recommendations for the redesign of financial incentives

Identified Gaps	Background/Field Findings	Researchers' Recommendations
The need for economic incentives	Farmers require economic support to implement and develop PRLB. The economic benefits of PRLB have not yet been felt equally across the board, whilst PRLB generates environmental benefits for which no economic compensation has yet been provided.	Green credit, price premium, PES, and carbon incentives
Risk protection needs to be strengthened during the implementation of the PRLB	Farmers face production and income risks during the implementation of, or transition to, PRLB practices.	Green insurance
The need to strengthen support for organic fertilizer inputs	Subsidies for organic fertilizers remain low.	Organic fertilizer subsidy

Note: PRLB = Sustainable Environmentally Friendly Agriculture (Pertanian Ramah Lingkungan Berkelanjutan); PES = Payment for Environmental Services.

Economic benefits can also be enhanced through a premium pricing mechanism for products produced using environmentally friendly agricultural practices. The price difference between organic and conventional products can serve as an incentive for producers to maintain these production systems (Pawlewicz, 2020). This mechanism can be implemented through collaboration between farmers, cooperatives, and marketing agencies that offer a price premium above market rates for PRLB products. In addition to generating economic value, implementing PRLB also provides environmental benefits that deserve recognition. Marketing agencies under local government management must be strengthened through capacity-building to compete effectively in marketing PRLB paddy or rice.

The implementation of PRLB generates environmental benefits that can form the basis for providing economic incentives through Payment for Environmental Services (PES) and carbon incentives. Improved soil quality, reduced chemical residues, and enhanced agricultural ecosystems are benefits resulting from implementing PRLB. Recognition of environmental services is increasingly being used as an instrument to promote sustainable resource management (Ibrahim et al., 2024). Therefore, the PES scheme and carbon incentives can be implemented as forms of compensation for the environmental benefits generated by farmers. The PES approach is considered effective for protecting agroforestry systems while providing additional benefits to smallholder farmers (Ibrahim et al., 2024). The implementation of PES is expected to succeed in PRLB, although the success of PRLB implementation remains influenced by farmers' ability to cope with various agricultural business risks (Ibrahim et al., 2024).

From a risk management perspective, existing agricultural insurance subsidies can be developed into green insurance that specifically supports PRLB farmers. A green insurance scheme could be implemented through larger premium subsidies for PRLB farmers or by providing coverage during the transition period toward an environmentally friendly farming system. Such a green insurance scheme could provide significant benefits to farmers during the transition from conventional to environmentally friendly agriculture (Hou & Wang, 2023).

3.5 Non-Financial Incentive Redesign

PRLB development also requires non-financial incentives that can strengthen farmers' capacity and support their access to markets. Research findings indicate that farmers still face limitations in market information and marketing networks and require capacity building not only in technical aspects of cultivation but also in managerial and business development skills. Recommendations for the redesign of non-financial incentives based on field findings are presented in Table 7.

Based on Table 7, one form of non-financial incentive that needs to be strengthened is capacity building for farmers. The development of environmentally friendly agriculture also requires non-financial incentives to build farmers' capacity (Yanfika et al., 2024). Ongoing technical assistance needs to be strengthened to ensure that farmers receive support throughout the implementation of sustainable agriculture. Technical assistance can be provided through collaboration among extension workers, universities, and farmer groups, covering topics such as crop cultivation, business management, marketing, and risk management. This capacity-building effort must be followed by recognition of the competencies that farmers have acquired.

One way of recognizing such capacity building is through the certification of farmers' competencies. Competency certification can serve as recognition of farmers' ability to apply environmentally friendly cultivation

standards. Such recognition can enhance stakeholders' trust in the quality of the production processes being implemented (Leitner & Vogl, 2020). Certification is granted based on training and competency evaluations, thereby serving as proof of farmers' ability to apply PRLB principles. In addition to providing formal recognition, farmers' motivation must also be continuously strengthened.

Table 7. Recommendations for the redesign of non-financial incentives

Identified Gaps	Background/Field Findings	Researchers' Recommendations
The need for information on market access	Farmers still face a lack of market information and the marketing networks needed to boost the uptake of PRLB products.	Market information system dan partnership arrangements
Farmers need managerial skills	Farmers need not only technical skills, but also managerial skills, business development, marketing and capacity-building to meet market demands.	Capacity-building, mentoring, competency certification, and awards/recognition

Note: PRLB = Sustainable Environmentally Friendly Agriculture (Pertanian Ramah Lingkungan Berkelanjutan).

Farmers' motivation can be strengthened by awarding prizes or recognition for the successful implementation of PRLB. Awarding recognition to farmers, farmer groups, or villages that have successfully and consistently implemented PRLB can serve as a form of appreciation for their achievements. Social recognition is often used to increase participation and commitment in the implementation of a development program (Jütersonke & Groß, 2023). Awards can be given through special incentive programs for high-performing participants (McDonagh, 2022). These efforts need to be supported by adequate access to information so that the benefits obtained can be more widely disseminated.

Strengthening farmers' capacity must be accompanied by the provision of market information that is easily accessible and regularly updated. The availability of information regarding prices, demand, marketing opportunities, and consumer preferences is a critical factor in agricultural decision-making. Access to relevant information can help reduce uncertainty in economic activities (Nagar et al., 2019). Information can be provided through digital platforms or market information systems that farmers can access regularly. The available information will be more effective if supported by a strong collaborative network.

To ensure that market information translates into broader market access, partnership networks among farmers, cooperatives, business actors, marketing institutions, and the processing industry should be strengthened as part of the PRLB development strategy. Collaboration among actors in the agricultural value chain can create more stable and mutually beneficial business relationships (Zhao et al., 2023). Partnerships can be established through marketing cooperation agreements, the procurement of agricultural produce, or the development of specialised markets for PRLB products, thereby better ensuring the sustainability of farming operations.

3.6 Farmer Capacity-Building Framework

The farmer capacity-building framework proposed in this study consists of four main dimensions: technical, managerial, institutional, and digital capacity. These four dimensions complement one another in supporting the sustainable implementation of PRLB. Farmer capacity building is a critical factor in accelerating the adoption of innovations and strengthening farm performance (Rana et al., 2025).

Technical capacity focuses on farmers' ability to implement environmentally friendly farming practices, integrated pest management, and soil and water conservation. Mastery of technical aspects serves as the foundation for reducing dependence on chemical inputs and maintaining sustainable land productivity (Bhujel & Joshi, 2024). Adequate technical capacity is essential for the successful implementation of sustainable agricultural technologies and practices (Coulibaly et al., 2021).

Farmers need managerial support that enables more efficient management of their farming operations. Managerial skills include business planning, risk management, and farm record-keeping as a basis for decision-making (Taramuel-Taramuel et al., 2023). Managerial skills play a role in improving business efficiency and farmers' resilience to various agricultural risks (Taramuel-Taramuel et al., 2023).

The successful implementation of PRLB also requires strong institutional support. The roles of farmer groups, cooperatives, and business partnerships are crucial in expanding access to resources, strengthening bargaining power, and improving market access. Farmer institutions enhance access to information, markets, and various agricultural support services (Abebaw & Haile, 2013).

Technological advancements have also positioned digital capacity as a key component in PRLB development. The use of smart farming technologies, market information systems, and digital marketing is expected to improve business efficiency while expanding market reach (Gyamfi et al., 2024). Digital transformation offers opportunities to enhance productivity and competitiveness in the agricultural sector by enabling faster, more accurate access to information.

The integration of these four dimensions demonstrates that the development of PRLB requires not only incentive

support but also the enhancement of farmers' capabilities to manage their businesses, build networks, and utilize technology. Synergy between the redesign of incentives and capacity building is expected to accelerate the achievement of the Ngawi Go Organic 2030 program.

3.7 Adoption of Sustainable Environmentally Friendly Agriculture

The proposed capacity-building for farmers and the redesign of incentives are expected to encourage wider adoption of PRLB practices. However, this study does not empirically measure changes in adoption rates resulting from the proposed redesign of incentives. This adoption is an important indicator because it reflects a shift in farmers' behavior toward more sustainable farming systems. The rate of innovation adoption is one measure of the success of agricultural development programs (Mutolib et al., 2026). The expected changes are reflected in reduced use of synthetic fertilizers and pesticides, and increased use of organic fertilizers, biofertilizers, and botanical pesticides in farming activities. These potential changes would represent an initial step toward a more environmentally friendly production system.

The implementation of PRLB is also reflected in increased resource-conservation practices, particularly in soil and water management. These efforts are crucial for maintaining land productivity while reducing pressure on the agricultural environment. Resource conservation is a key component in ensuring the long-term sustainability of agricultural production systems (Sharma et al., 2023). Ultimately, these various changes are expected to lead to the establishment of a sustainable production system, one capable of maintaining a balance between productivity, environmental conservation, and farmer well-being. Sustainable production emphasizes a balance among economic, social, and environmental aspects in agricultural activities (Sargani et al., 2020). Thus, the adoption of PRLB practices is the ultimate desired outcome of integrating the redesign of financial incentives, the redesign of non-financial incentives, and capacity building for farmers. The higher the adoption rate, the greater the likelihood of achieving a sustainable and competitive agricultural system.

3.8 Long-Term Impacts of the Development of Sustainable Environmentally Friendly Agriculture

The redesign of financial and non-financial incentives, supported by capacity-building for farmers, is expected to accelerate the adoption of PRLB practices. However, this acceleration in adoption is a potential impact of the proposed recommendations and was not empirically measured in this study. The combination of incentives, capacity building, and institutional support is known to play a crucial role in promoting the adoption of sustainable agriculture (Rizzo et al., 2024). Enhanced farmer capacity, supported by institutional strengthening and business partnerships, will improve access to markets and marketing information (Permadi & Winarti, 2025). This could provide farmers with greater opportunities to secure higher market prices through the development of specialised markets, product certification, and more organized marketing collaborations. Market access, certification, and strong institutional frameworks contribute to increased value added in agricultural products (Zhang et al., 2024).

This higher value added is expected to provide more tangible economic benefits for farmers. In addition to increased product sales value, economic benefits can be strengthened through more efficient use of inputs and reduced farming risks. The economic benefits experienced by farmers are a key factor in ensuring the sustainable adoption of an agricultural innovation (Hanani AR et al., 2024). This situation is expected to encourage farmers to maintain and expand the application of PRLB practices.

The continued implementation of PRLB will further strengthen farmers' ability to produce products that meet market demands, which are increasingly focused on quality, food safety, and sustainability. Enhancing farmers' capacity, improving the quality of certified products, and strengthening institutional frameworks through expanded market access provide significant economic benefits and ensure the sustainability of the agroindustry (Yusuf et al., 2025). The increasingly widespread adoption of environmentally friendly practices will boost market confidence in the products produced, thereby opening up greater opportunities for business development (Lin et al., 2025).

In the long term, increased PRLB adoption not only provides economic benefits but also supports the environmental sustainability of agriculture. Reducing synthetic chemical inputs, increasing the use of organic materials, and implementing natural resource conservation are expected to sustainably preserve soil, water, and agroecosystem quality (Panwar et al., 2022). Overall, the redesign of incentives, supported by capacity-building for farmers, is expected to create a more sustainable PRLB system by simultaneously increasing economic benefits, strengthening product competitiveness, and improving environmental quality. These conditions are expected to support the achievement of the Ngawi Go Organic 2030 program and strengthen the development of environmentally friendly rice farming in Ngawi Regency. However, these impacts were not empirically tested in this study and should therefore be considered potential outcomes of the proposed incentive redesign.

4. Conclusions

This study formulates an incentive design to enhance the capacity of farmers practicing PRLB in Ngawi Regency, emphasizing the integration of redesigned financial and non-financial incentives and capacity-building

for farmers. The results show that, while the government has provided various incentives, gaps remain in marketing, economic benefits, risk management, the use of environmentally friendly inputs, and the development of farmers' business capacity. The proposed framework indicates that the redesign of financial incentives can be achieved through the development of organic fertilizer subsidies, green credit, price premiums, PES, and green insurance. Meanwhile, the redesign of non-financial incentives can be carried out through ongoing mentoring, competency certification, awards, the provision of market information, and the strengthening of partnerships. Capacity building for farmers focuses on strengthening technical, managerial, institutional, and digital capacities to support the wider adoption of PRLB practices. Overall, this framework provides a strategic framework to increase PRLB adoption, strengthen farmers' competitiveness and well-being, and support environmental sustainability in realizing the Ngawi Go Organic 2030 program.

Author Contributions

Conceptualization, H.I. and M.F.Y.P.; methodology, H.I. and H.N.; validation, B.W.U.; formal analysis, H.I., E.W.R., B.W.U., H.N., M.F.Y.P., and Y.S.A.; investigation, E.W.R. and M.F.Y.P.; data curation, E.W.R. and M.F.Y.P.; writing—original draft preparation, H.I.; writing—review and editing, Y.S.A.; supervision, E.W.R. and Y.S.A. All authors have read and agreed to the published version of the manuscript.

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Informed Consent Statement

Informed consent was obtained from all participants prior to their participation in the study.

Data Availability

The datasets generated and/or analyzed during the current study are available from the corresponding author upon reasonable request.

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Conflicts of Interest

The authors declare no conflicts of interest.

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